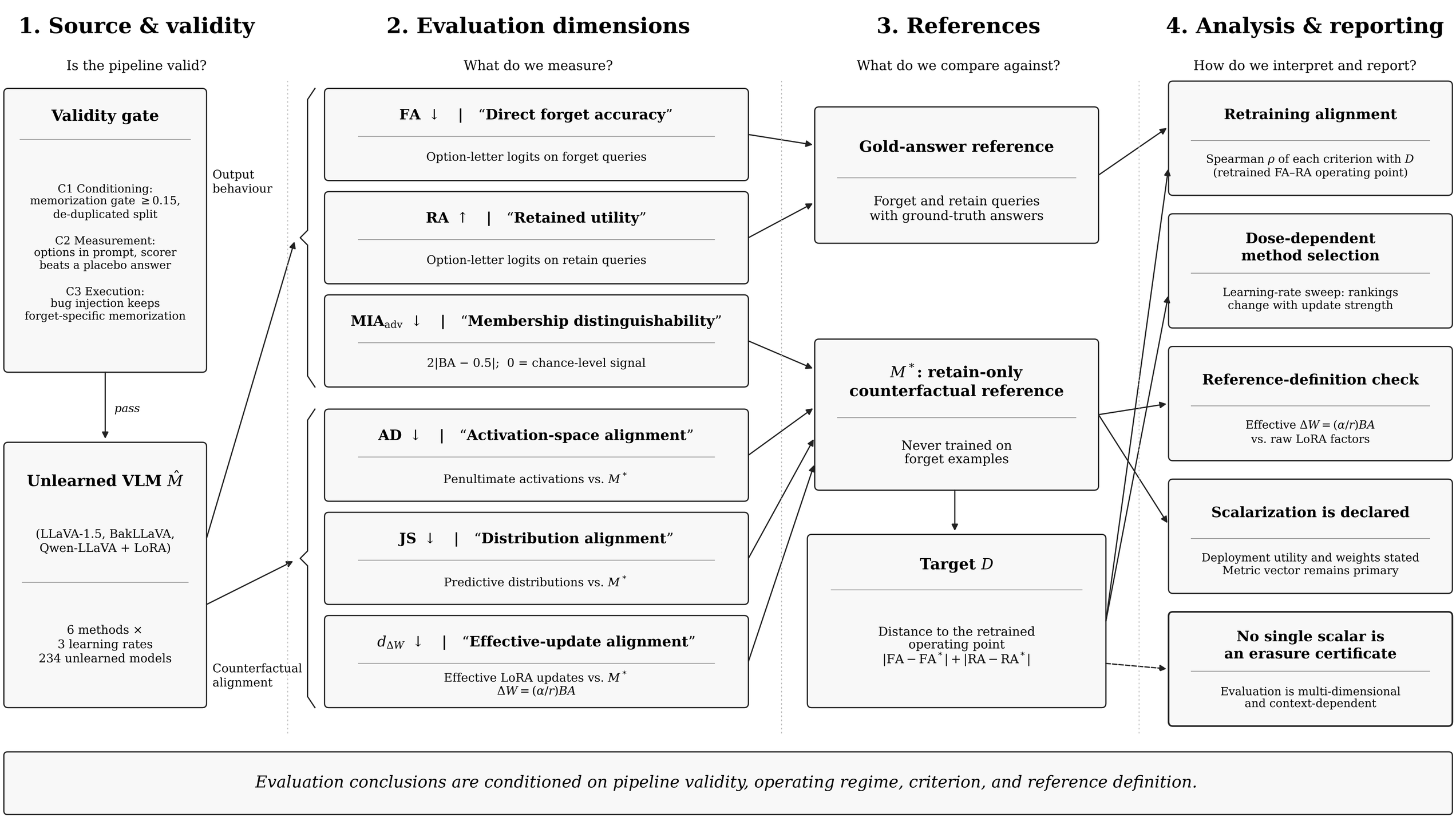

1. Source & validity
Is the pipeline valid?
Validity gate
C1 Conditioning:
memorization gate ≥ 0.15,
de-duplicated split
C2 Measurement:
options in prompt, scorer
beats a placebo answer
C3 Execution:
bug injection keeps
forget-specific memorization
pass
Unlearned VLM $\hat{M}$
(LLaVA-1.5, BakLLaVA,
Qwen-LLaVA + LoRA)
6 methods ×
3 learning rates
234 unlearned models
Output behaviour
Counterfactual alignment
2. Evaluation dimensions
What do we measure?
FA ↓ | "Direct forget accuracy"
Option-letter logits on forget queries
RA ↑ | "Retained utility"
Option-letter logits on retain queries
MIA$_{adv}$ ↓ | "Membership distinguishability"
2|BA − 0.5|; 0 = chance-level signal
AD ↓ | "Activation-space alignment"
Penultimate activations vs. M^*
JS ↓ | "Distribution alignment"
Predictive distributions vs. M^*
$d_{\Delta W}$ ↓ | "Effective-update alignment"
Effective LoRA updates vs. M^*
$\Delta W = (\alpha/r)BA$
3. References
What do we compare against?
Gold-answer reference
Forget and retain queries
with ground-truth answers
M^*: retain-only counterfactual reference
Never trained on
forget examples
Target D
Distance to the retrained
operating point
$|FA - FA^*| + |RA - RA^*|$
4. Analysis & reporting
How do we interpret and report?
Retraining alignment
Spearman ρ of each criterion with D
(retrained FA–RA operating point)
Dose-dependent method selection
Learning-rate sweep: rankings
change with update strength
Reference-definition check
Effective $\Delta W = (\alpha/r)BA$
vs. raw LoRA factors
Scalarization is declared
Deployment utility and weights stated
Metric vector remains primary
No single scalar is an erasure certificate
Evaluation is multi-dimensional
and context-dependent
Evaluation conclusions are conditioned on pipeline validity, operating regime, criterion, and reference definition.